Calculus II

Solution to Assignment 9

Part I. Regular Problems

1. Compute the partial derivatives.

   (a) Find f_{xx} and f_{zz} for $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$, $(x,y,z) \neq (0,0,0)$.
   First,
   $$f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$$
   Thus,
   $$f_x = \frac{d}{dx}\left(x^2+y^2+z^2\right)^{-1/2} = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2x) = -x(x^2+y^2+z^2)^{-3/2}$$
   Then,
   $$f_{xx} = \frac{\partial}{\partial x}\left(-x(x^2+y^2+z^2)^{-3/2}\right)$$
   Using product rule:
   $$f_{xx} = -(x^2+y^2+z^2)^{-3/2} + x\left(\frac{3}{2}\right)(x^2+y^2+z^2)^{-5/2}(2x)$$
   Simplify:
   $$f_{xx} = -(x^2+y^2+z^2)^{-3/2} + 3x^2(x^2+y^2+z^2)^{-5/2}$$
   Similarly,
   $$f_z = -z(x^2+y^2+z^2)^{-3/2}$$
   thus
   $$f_{zz} = -(x^2+y^2+z^2)^{-3/2} + 3z^2(x^2+y^2+z^2)^{-5/2}$$

   (b) Find f_{xyz} if $f(x,y,z) = xy + yz + zx$.
   First,
   $$f_x = y+z, \quad f_{xy} = \frac{\partial}{\partial y}(y+z) = 1, \quad f_{xyz} = \frac{\partial}{\partial z}(1) = 0$$
   Thus,
   $$f_{xyz} = 0$$

(c) Find $g_{xxy} - g_{xyx}$ if g is C^3 (three times continuously differentiable).
By Clairaut's theorem (equality of mixed partials):

$$g_{xxy} = g_{xyx}$$

Thus:

$$g_{xxy} - g_{xyx} = 0$$

(d) Find h_{x_j} and $h_{x_j x_k}$ for $h(\mathbf{X}) = \mathbf{A} \cdot \mathbf{X}$.
We have:

$$h(\mathbf{X}) = \sum_{i=1}^{n} a_i x_i$$

Thus,

$$h_{x_j} = a_j, \quad h_{x_j x_k} = 0 \quad \text{(since derivatives of constants are zero)}$$

2. Let $f(x, y, z) = x^2 + xy + \frac{1}{2}y^2 + 2yz + z^3$.

(a) Compute ∇f:

$$\nabla f = (2x + y, x + y + 2z, 2y + 3z^2)$$

Setting $\nabla f = \mathbf{0}$:

$$\begin{cases} 2x + y = 0 \\ x + y + 2z = 0 \\ 2y + 3z^2 = 0 \end{cases}$$

From the first, $y = -2x$. Substitute into second:

$$x - 2x + 2z = 0 \Rightarrow -x + 2z = 0 \Rightarrow x = 2z$$

Substituting $x = 2z$ into $y = -2x$:

$$y = -4z$$

Substituting into third equation:

$$2(-4z) + 3z^2 = 0 \quad \Rightarrow \quad -8z + 3z^2 = 0 \quad \Rightarrow \quad z(3z - 8) = 0$$

Thus, $z = 0$ or $z = \frac{8}{3}$.
For $z = 0$, $(x, y, z) = (0, 0, 0)$.
For $z = 8/3$:

$$x = 2(8/3) = 16/3, \quad y = -4(8/3) = -32/3$$

Thus, $\mathbf{A} = \left(\frac{16}{3}, -\frac{32}{3}, \frac{8}{3}\right)$.

(b) Show local minimum at $\mathbf{A}$.

The Hessian matrix is:

$$H = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 6z \end{pmatrix}$$

At $\mathbf{A}$, $z = 8/3$, so $6z = 16$.

Thus,

$$H = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 16 \end{pmatrix}$$

We can check $2 > 0$, $2 - 1 = 1 > 0$ and $2 \times (16 - 4) - 16 = 8 > 0$, so H is positive definite $\Rightarrow$ local minimum at $\mathbf{A}$.

(c) Find points near $(0, 0, 0)$ where $f > 0$ and $f < 0$.

At $(0.1, 0, 0)$:

$$f(0.1, 0, 0) = (0.1)^2 = 0.01 > 0$$

At $(0, 0.1, 0)$:

$$f(0, 0.1, 0) = \frac{1}{2}(0.1)^2 = 0.005 > 0$$

At $(0, 0, -0.1)$:

$$f(0, 0, -0.1) = (-0.1)^3 = -0.001 < 0$$

Near $(0, 0, 0)$, f takes both positive and negative values, which indicates $(0, 0, 0)$ is a saddle point.

3. Find the maximum of $Q(x, y) = 3x^2 + 2xy + 3y^2$ on $x^2 + y^2 = 1$.

Use Lagrange multipliers:

$$\nabla Q = (6x + 2y, 2x + 6y), \quad \nabla g = (2x, 2y)$$

$$(6 - 2\lambda)x + 2y = 0, \quad 2x + (6 - 2\lambda)y = 0, \quad x^2 + y^2 = 1.$$

We get $y = (-3 + \lambda)x, x + (3 - \lambda)(-3 + \lambda)x = 0$, so $x = 0$ or $1 - (\lambda - 3)^2 = 0$. So $\lambda = 2$ or 4. Using $4, y = x$ and $Q\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (3 + 2 + 3)\frac{1}{2} = 4 = Q\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$. That is the maximum because using $\lambda = 2$ you find $y = -x$ and $Q\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = (3 - 2 + 3)\frac{1}{2} = 2 = Q\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. Thus, maximum value is 4.

4. Evaluate integrals over $R = [0, 1] \times [0, 1]$:

(a)

$$\begin{aligned}\iint_R (x^3+y^2)dA &= \int_0^1\int_0^1 x^3 dxdy + \int_0^1\int_0^1 y^2 dxdy \\ &= \int_0^1 \frac{1}{4}dy + \int_0^1 y^2 dy \\ &= \frac{1}{4}+\frac{1}{3} = \frac{7}{12}\end{aligned}$$

(b)

$$\begin{aligned}\iint_R ye^{xy}dA &= \int_0^1 \left(\int_0^1 ye^{xy}dx\right)dy \\ &= \int_0^1 \left(e^{xy}\mid_{x=0}^2\right)dy \\ &= \int_0^1 (e^y-1)dy \\ &= (e-2)\end{aligned}$$

(c)

$$\begin{aligned}\iint_R (xy)^2\cos(x^3)dA &= \int_0^1 \left(\int_0^1 x^2y^2\cos(x^3)dy\right)dx \\ &= \int_0^1 \frac{1}{3}x^2\cos(x^3)dx \qquad [\text{Substitution } u=x^3,\ du=3x^2dx] \\ &= \frac{1}{9}\int_0^1 \cos(u)du \\ &= \frac{1}{9}(\sin(1)-\sin(0)) = \frac{\sin(1)}{9}\end{aligned}$$

(d)

$$\iint_R \log[(x+1)(y+1)]dA$$

Expand:

$$= \int_0^1\int_0^1 (\log(x+1)+\log(y+1))\,dA$$

Separates into two integrals:

$$= \left(\int_0^1\int_0^1 \log(x+1)dydx\right)+\left(\int_0^1\int_0^1 \log(y+1)dxdy\right) = \int_0^1 \log(x+1)dx+\int_0^1 \log(y+1)dy$$

They are equal, so:

$$= 2\times\int_0^1 \log(x+1)dx$$

Use integration by parts:

$$\begin{aligned}\int_0^1 \log(x+1)dx &= x\log(1+x)\,|_0^1 - \int_0^1 x d\left(\log(1+x)\right)\\ &= \log 2 - \int_0^1 \frac{x}{1+1}dx\\ &= \log 2 - \int_0^1 \frac{(1+x)-1}{1+x}dx\\ &= \log 2 - \left(\int_0^1 1dx - \int_0^1 \frac{1}{1+x}dx\right)\\ &= \log 2 - (1-\log 2)\\ &= 2\log 2 - 1\end{aligned}$$

Thus $\iint_R \log[(x+1)(y+1)]dA = 4\log 2 - 2$

5. Evaluate $\iint_R f(x,y)dxdy$ for given functions:

   (a) $f(x,y) = (x+2y)^2$ over $R = [-1,2]\times[0,2]$.
   Expand:

$$(x+2y)^2 = x^2 + 4xy + 4y^2$$

   Integrate term by term:

$$\begin{aligned}\int_{-1}^2\int_0^2 (x+2y)^2 dydx &= \int_{-1}^2\int_0^2 (x^2+4xy+4y^2)dydx\\ &= \int_{-1}^2\int_0^2 x^2 dydx + \int_{-1}^2\int_0^2 4xydydx + \int_{-1}^2\int_0^2 4y^2 dydx\\ &= 6 + 12 + 32\\ &= 50\end{aligned}$$

   (b) $f(x,y) = y^3\cos^2 x$ over $R = [-\pi/2,\pi]\times[1,2]$.
   Use iterated integral:

$$\int_1^2\int_{-\pi/2}^{\pi} y^3\cos^2 x\,dxdy$$

   First integral:

$$\int_{-\pi/2}^{\pi} y^3\cos^2 x\,dx = y^3\int_{-\pi/2}^{\pi}\frac{1+\cos(2x)}{2}\,dx = \frac{y^3}{2}\int_{-\pi/2}^{\pi} 1\,dx + \frac{y^3}{2}\int_{-\pi/2}^{\pi}\cos(2x)\,dx$$

$$= \frac{y^3}{2}x\,|_{-\pi/2}^{\pi} + \frac{y^3}{4}\sin(2x)\,|_{-\pi/2}^{\pi} = \frac{y^3}{2}(\pi+\frac{\pi}{2}) + \frac{y^3}{4}(0-0) = \frac{3\pi y^3}{4}$$

Second integral:

$$\int_1^2 \frac{3\pi y^3}{4}\,dy = \frac{3y^4}{16}\,|_1^2 = \frac{45\pi}{16}$$

Thus, the result is $\frac{45\pi}{16}$.

(c) $f(x,y) = xy^3e^{x^2y^2}$ over $R = [1,3]\times[1,2]$.
Use substitution $u = x^2y^2$. Then

$$\begin{aligned}\int_1^3\int_1^2 xy^3e^{x^2y^2}dy\,dx &= \int_1^2 \frac{y}{2}\left(\int_{1^2\cdot y^2}^{3^2\cdot y^2} e^u du\right)\,dy\\ &= \int_1^2 \frac{y}{2}(e^{9y^2}-e^{y^2})dy\\ &= \int_1^2 \frac{1}{36}e^{9y^2}dy^2 - \int_1^2 \frac{1}{4}e^{y^2}dy^2\\ &= \frac{1}{36}(e^{36}-e^9) - \frac{1}{4}(e^4-e)\end{aligned}$$

(d) $f(x,y) = xy + \frac{x}{y+1}$ over $R = [1,4]\times[1,2]$.
Integrate term-by-term:

$$\begin{aligned}\int_1^4\int_1^2 xy + \frac{x}{y+1}dy\,dx &= \int_1^4\int_1^2 xydy\,dx + \int_1^4\int_1^2 \frac{x}{y+1}dy\,dx\\ &= \int_1^4 x\left(\frac{y^2}{2}\,|_1^2\right)dx + \int_1^4 x\left(\ln(1+y)\,|_1^2\right)dx\\ &= \frac{3x^2}{4}\,|_1^4 + \frac{x^2\ln(\frac{3}{2})}{2}\,|_1^4\\ &= \frac{45}{4} + \frac{15}{2}\ln(\frac{3}{2})\end{aligned}$$

Part II. Group Problems

1. Let $f(x,y) = (x-1)^2 + 2(y-2)^2 + (y-2)^3$.
   Expand:

$$f(x,y) = 1 - 2x + x^2 + 4y - 4y^2 + y^3$$

   (a) Gradient:

$$\nabla f = (2x-2, 4-8y+3y^2)$$

   Set $\nabla f = 0$:

$$\begin{aligned}2x-2=0 \quad &\Rightarrow \quad x=1\\ 4-8y+3y^2=0 \quad &\Rightarrow y=2 \text{ or } y=\frac{2}{3}\end{aligned}$$

   So the two points are $(1,2)$ and $(1,\frac{2}{3})$

(b) Hessian matrix:

$$\mathcal{H}f = \begin{pmatrix} 2 & 0 \\ 0 & -8+6y \end{pmatrix}$$

Thus

$$\mathcal{H}f(1,2) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \quad \mathcal{H}f(1,\frac{2}{3}) = \begin{bmatrix} 2 & 0 \\ 0 & -4 \end{bmatrix}$$

(c) Since $2 > 0$ and $2 \times 4 - 0^2 = 8 > 0$, $\mathcal{H}f(1,2)$ is positive definite, then f has a local minimum at $(1,2)$. f has a saddle at $(1, \frac{2}{3})$ because $2 \times (-4) - 0^2 = -8 < 0$.

2. Evaluate double integrals:

(a)

$$\iint_R 7\, dA = 7 \times (5-3) \times (9-(-1)) = 7 \times 2 \times 10 = 140$$

$$\iint_R x\, dA = \int_0^1 \int_0^1 x\, dy\, dx = \int_0^1 x\, dx = \frac{1}{2}$$

(b)

$$\begin{aligned} \iint_R (x^2y^2 + x)\, dy\, dx &= \int_0^2 \left(\int_{-1}^0 x^2y^2\ dy + \int_{-1}^0 x\ dy \right) dx \\ &= \int_0^2 \left(\frac{x^2}{3}y^3\ |_{-1}^0 + xy\ |_{-1}^0 \right) dx \\ &= \int_0^2 \left(\frac{x^2}{3} + x \right)\ dx \\ &= \frac{x^3}{9}\ |_0^2 + \frac{x^2}{2}\ |_0^2 \\ &= \frac{26}{9} \end{aligned}$$

(c)

$$\begin{aligned} \iint_R (|y| \cos(\frac{\pi}{4}x))\, dy\, dx &= \int_0^2 \left(\int_{-1}^0 \cos(\frac{\pi}{4}x)\, (-y)\ \ dy \right) dx \\ &= \int_0^2 \cos(\frac{\pi}{4}x) \cdot \frac{1}{2} dx = \frac{2}{\pi} \sin(\frac{\pi}{4}x)\ |_0^2 \\ &= \frac{2}{\pi} \end{aligned}$$